Fabrication and Test of Nb_3Sn-NbTi Superconducting Joint for a 14-T MRI Magnet

Zhidong Shen, Xiaohua Jiang, Guolin Chai, Zhan Zhang, Peng Gao, Chao Zhou, and Ye Li

***Abstract*—The 14-T magnetic resonance imaging (MRI) superconducting magnet utilizes reinforced multifilament Nb_3Sn wires and multifilament NbTi wires to generate the required magnetic field. The resistances of the Nb_3Sn-NbTi superconducting joints are crucial to the temporal field stability of the magnet operation in persistent mode. The resistance of each joint is required to be lower than 10^{-11} Ω under a 0.5 T background magnetic field to keep the temporal field drift lower than 0.1 ppm/hour. This paper presents the fabrication of multifilament Nb_3Sn-NbTi superconducting joints and the measurement of their resistances. The joints were fabricated by the solder matrix replacement method using PbBi as the solder in an open-air environment. The samples were tested by an apparatus that was developed based on the current-decay method and can provide a background field up to 1 T. The sample loop with a Nb_3Sn wire and a NbTi wire contains two Nb_3Sn-NbTi joints. The Nb_3Sn wire undergoes the heat treatment before the fabrication of superconducting joints with the NbTi wire. The measured results indicate that the joints can carry a current up to 809.8 A with a resistance lower than 6.20×10^{-13} Ω without a background field. Under a 1 T background field, the joints can carry a current up to 481.5 A with a resistance lower than 1.24×10^{-12} Ω, which meet the requirement of the 14-T MRI superconducting magnet.**

***Index Terms*—Current-decay method, high-field magnetic resonance imaging (MRI) magnet, Nb_3Sn, NbTi, superconducting joint.**

I. Introduction

THE 14-T magnetic resonance imaging (MRI) superconducting magnet consists of multiple superconducting solenoids [1], [2]. Considering that Nb_3Sn wires exhibit superior critical current under high magnetic fields and NbTi wires exhibit lower cost [3], both types of superconducting wires were utilized in the design of the 14-T MRI magnet. Due to constraints imposed by the manufacturing techniques of superconducting wires, the maximum length of a single superconducting wire is limited [3]. Furthermore, connections between Nb_3Sn and NbTi coils are required. Therefore, there are multiple superconducting joints in the 14-T MRI magnet operating in the persistent mode. The resistance of the superconducting joints constitutes the major component of the total resistance in the magnet. In order to keep the temporal drift of the main magnetic field lower than 0.1 ppm/hour, it is necessary to minimize the resistance of the superconducting joints. In the 14-T MRI magnet, the superconducting joints are located away from the high-field region [1]. Under the magnetic field of 0.5 T at the joint location, the resistance of each joint is required to be lower than 10^{-11} Ω.

There are two typical methods for fabricating Nb_3Sn-NbTi superconducting joints: solder matrix replacement method [4], [5] and the jointing method utilizing superconducting alloy tube as the intermedium [6], [7]. The solder matrix replacement method is relatively simple and reliable [5], [8]. The Nb_3Sn-NbTi joints have been fabricated by solder matrix replacement and tested by four-probe technique by [4]. However, the four-probe method can not measure resistance lower than 10^{-11} Ω [4]. It is necessary to employ the current-decay method with higher precision to verify the reliability of using the solder matrix replacement method for fabricating Nb_3Sn-NbTi superconducting joints.

This paper presents the fabrication of Nb_3Sn-NbTi superconducting joints by the solder matrix replacement method using PbBi as the solder in an open-air environment. A closed-loop sample ring containing two Nb_3Sn-NbTi superconducting joints was designed and fabricated. The Nb_3Sn wire must undergo the heat treatment before the fabrication of superconducting joints with the NbTi wire. Furthermore, an apparatus was developed based on the current-decay method to measure the critical current and the resistance of the sample joints in order to validate the joint fabrication process. The apparatus can supply a background magnetic field up to 1 T.

II. Testing Apparatus

The superconducting joint testing apparatus was housed in a cryogenic storage Dewar containing liquid helium. The main components of the apparatus are shown in Fig. 1. An AMI

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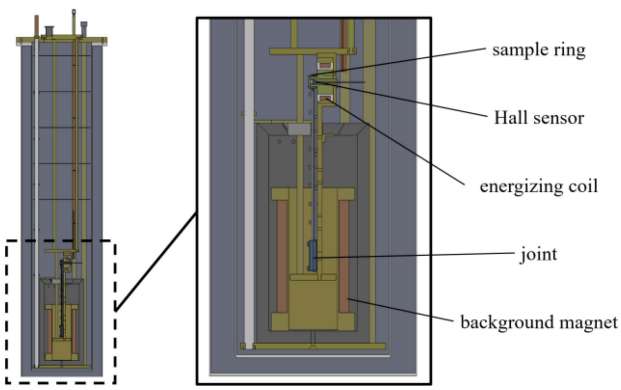

Fig. 1. Structure diagram of the testing apparatus for joints.

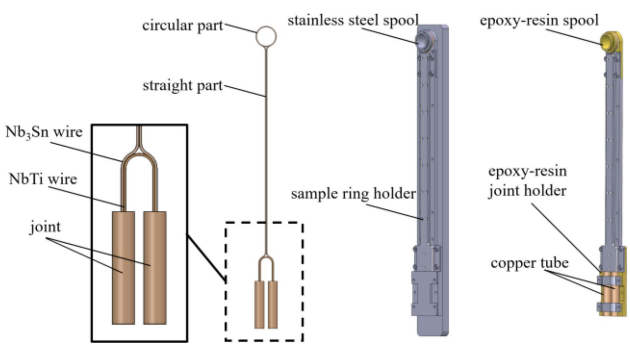

Fig. 2. Structure diagram of the sample ring and the holder.

standard rigid 4.2 K liquid helium level sensor was used to guarantee that the sample ring, the energizing coil and the background magnet were immersed in liquid helium and worked at 4.2 K.

The sample ring for current-decay measurement was required to form a closed-loop circuit. By measuring the time constant of this circuit, the resistance within the sample ring can be calculated by:

$$R = L\frac{ln\,(I_1/I_2)}{t_2 - t_1} \tag{1}$$

where L is the self-inductance of the sample ring, I is the current in the circuit. The sample ring wound with a Nb_3Sn wire and a NbTi wire contains two Nb_3Sn-NbTi superconducting joints. The self-inductance of the sample ring was calculated to be 0.20 μH using finite element simulation. As shown in Fig. 2, the sample ring consists of three parts: the circular part, the straight part and the joints.

The circular part was wound with the Nb_3Sn wire for two turns with an inner diameter of 20 mm. The energizing coil wound with another NbTi wire was positioned coaxially with the circular part of the sample ring, as shown in Fig. 3. The energizing coil has an exciting ratio to the sample ring current of around 1:5.88. A current of 170 A was applied to the energizing coil, inducing a current of 1000 A in the sample ring. A heater with power of 18 W was attached to the sample ring for trapping the magnetic field inside the circular part. A LakeShore HGCA-3020 Hall sensor was installed at the center of the circular part

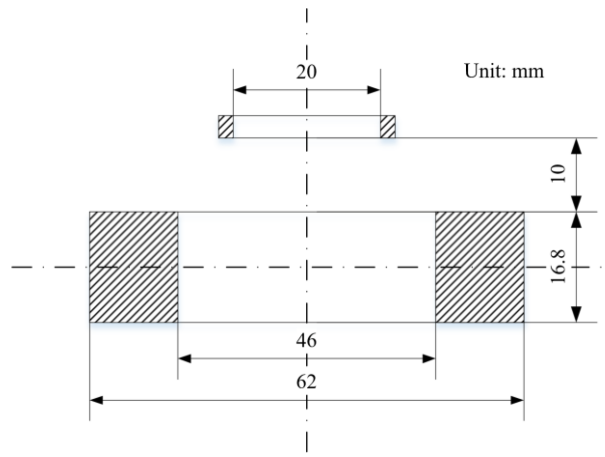

Fig. 3. Cross section scheme of the energizing coil and the sample ring.

of the sample ring to measure the magnetic field generated by the trapped current in the sample ring. Even though the number of turns is only two, the sample ring achieves a high ratio of 1111 G/kA between the generated central magnetic field and the trapped current due to its small inner diameter, which guarantees the measurement accuracy of the Hall sensor.

The length of the straight part is 210 mm, keeping the circular part away from the background magnetic field to avoid interference while placing the joints at the center of the background magnet. The background magnet wound with a NbTi wire can provide a 1 T magnetic field with a uniformity of 96.0 % within the central region of φ40 mm $\times$ 60 mm when energized to 85 A. The background magnet was surrounded by an iron shield and the background field was perpendicular to the direction of the Hall sensor. The calculated leakage magnetic field at the location of the Hall sensor is only 1.3 G, which is small enough not to disturb the measurement, indicating effective magnetic field shielding.

The sample ring holder was designed and fabricated accommodating the shape of the sample ring, as shown in Fig. 3. The sample ring holder entirely made of 304 stainless steel was used to provide support for the Nb_3Sn wire during the heat treatment process before joint fabrication. The joint holder is removable so that the ends of the Nb_3Sn and NbTi wires could be exposed for joint fabrication. After joint fabrication, the epoxy-resin joint holder was utilized to achieve insulation between the copper tubes and the sampler ring holder. During the current-decay measurement, the stainless steel spool was replaced with the epoxy-resin spool to avoid the magnetic shielding effect caused by eddy current in metallic components which would prevent the "trapping" of current in the sample ring.

III. Joint Fabrication

A. Heat Treatment

The main parameters of the superconducting wires used for fabricating the sample ring are listed in Table I. Before joint fabrication, the Nb_3Sn wire was wound on the stainless steel spool and heat treated in an argon atmosphere following standard procedure:

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TABLE I
MAIN PARAMETERS OF THE SUPERCONDUCTING WIRES USED FOR FABRICATING THE SAMPLE RING

	Cross-section Area (mm)	Ratio of Copper to Other Components	Material of the Reinforcer
Nb_3Sn wire	1.35×0.90	0.27	NbTi-Cu
NbTi wire	1.15×0.75	3.00	

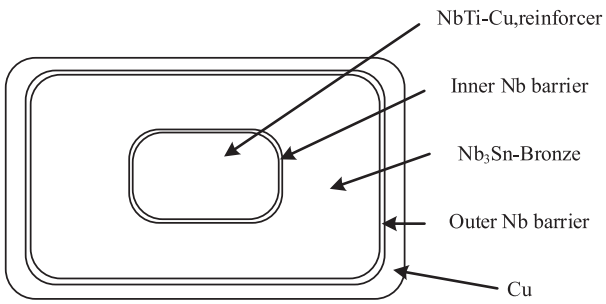

Fig. 4. Cross section scheme of the Nb_3Sn wire used for fabricating the sample ring.

1) ramp from room temperature to 670 °C at a rate of 30 °C/hour;
2) hold at 670 °C for 96 hours;
3) cool down from 670 °C to room temperature at a rate of 10 °C/hour.

B. Chemical Etching

The joint holder was detached to expose the ends of the Nb_3Sn wire and the braided E-glass fiber electrical insulation at the ends of the wire was removed for 50 mm. The cross section scheme of the Nb_3Sn wire is shown in Fig. 4. The copper matrix of the Nb_3Sn wire was chemically etched away by a concentrated nitric acid solution with a mass fraction of 68 %. Then the outer Nb barrier of the Nb_3Sn wire was etched away by a mixed acid solution of water, hydrofluoric acid, nitric acid and hydrochloric acid in a mass ratio of 75 : 5 : 10 : 10.

C. Solder Matrix Replacement

The Formvar electrical insulation at the ends of the NbTi wire was removed for 50 mm. The ends of the Nb_3Sn and the NbTi wires were joined side by side and immersed in molten Sn at 350 °C for 2 hours. During the process, an ultrasonic generator was immersed in the molten Sn to facilitate the replacing of the bronze matrix on the Nb_3Sn wire and the copper matrix on the NbTi wire with the molten Sn. Then, the ends of the Nb_3Sn and NbTi wires were immersed in molten PbBi solder at 350 °C for 30 minutes. During the process, an ultrasonic generator was immersed in the molten PbBi solder to facilitate the replacing of the Sn on the surfaces of Nb_3Sn and NbTi filaments with the PbBi solder, resulting in a superconducting connection between Nb_3Sn and NbTi. During the entire solder matrix replacement process, it is necessary to clear the surface oxide layer of the molten Sn and PbBi solder every 10 minutes. The superconducting wires, which had formed the superconducting connection, were immersed in oxygen-free copper tubes filled with molten PbBi solder at 140∼160 °C. The superconducting joints were formed after the cooling of the copper tubes.

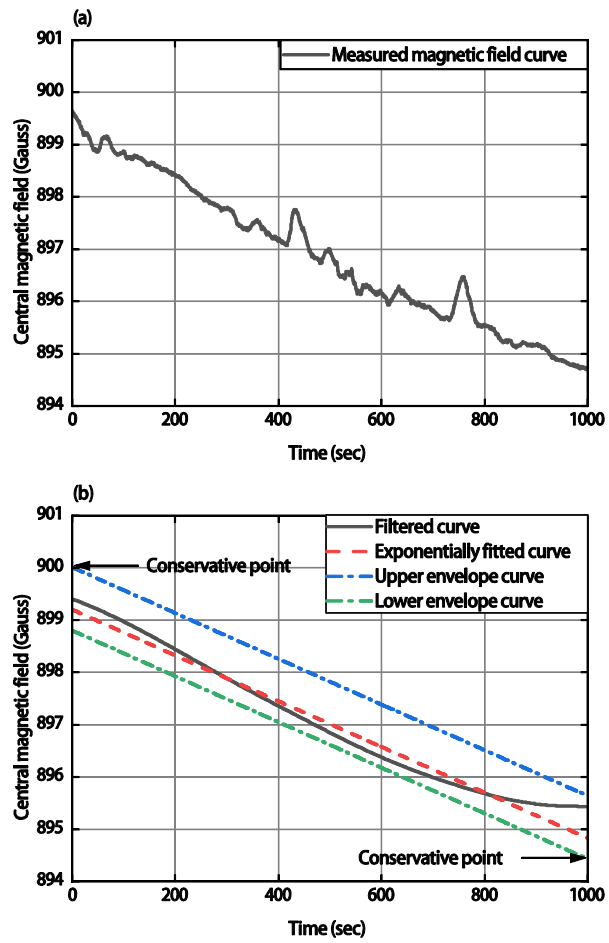

Fig. 5. Magnetic field decay curve at the center of the sample ring without background field.

IV. TEST RESULTS

The critical currents and the resistances of the Nb_3Sn-NbTi superconducting joints were tested once without background magnetic field and once under a field of 1 T. Figs. 5 and 6 show the magnetic field decay curves measured at the center of the sample ring.

The curve measured without background was low-pass filtered to eliminate the irregular fluctuations caused by temperature drift and electromagnetic interference and subsequently exponentially fitted. The obtained exponential curve was shifted upwards and downwards to form an envelope around the filtered curve. The two envelope curves were further shifted separately upwards and downwards by 0.20 G, which was the discrepancy between the zero drifts of the Hall sensor before and after the measurement. As a result, the true magnetic field decay curve was encompassed within the band region formed by the shifted

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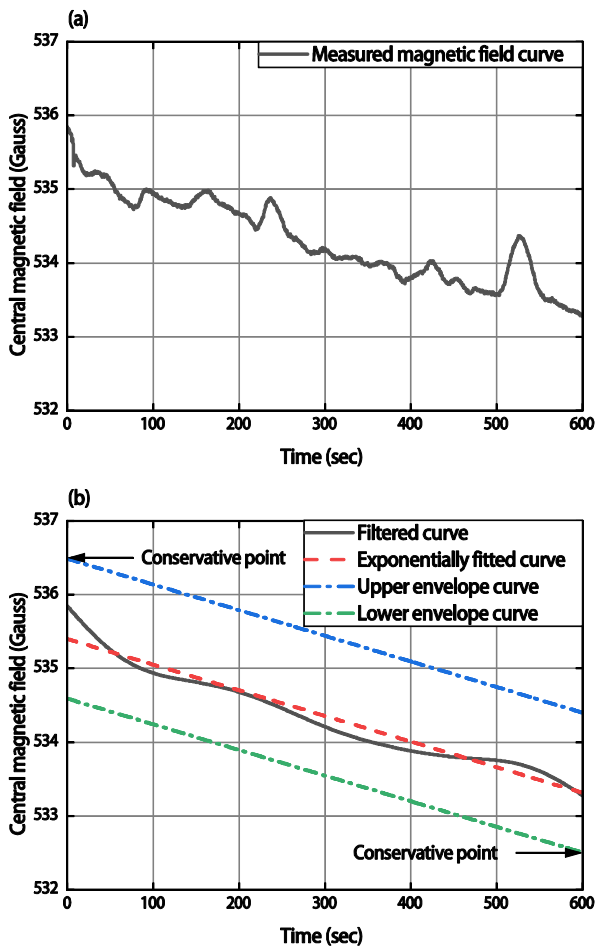

Fig. 6. Magnetic field decay curve at the center of the sample ring under a 1 T background field.

exponential curves. The conservative upper limit value for the resistance of the joints was obtained by calculating the time constant using the diagonal endpoints of the band region.

The same data processing method was applied to the analysis of the curve measured under 1 T background field.

The obtained results show that the joints can carry a current up to 809.8 A with a resistance lower than 6.20×10^{-13} Ω without background field. Under 1 T background field, the joints can carry a current up to 481.5 A with a resistance lower than 1.24×10^{-12} Ω.

V. Conclusion

A superconducting joint testing apparatus was developed based on the current-decay method, which can supply a background magnetic field up to 1 T. The Nb_3Sn-NbTi superconducting joints were fabricated by the solder matrix replacement method and tested using the apparatus. The measured results indicate that the joints can carry a current up to 809.8 A with a resistance lower than 6.20×10^{-13} Ω without a background field. Under a 1 T background field, the joints can carry a current up to 481.5 A with a resistance lower than 1.24×10^{-12} Ω. It indicates that the performance of the Nb_3Sn-NbTi superconducting joints meets the requirements of the 14-T MRI magnet, guaranteeing the temporal field drift lower than 0.1 ppm/hour.

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